

NOXUS — WEEK 7 & WEEK 8

WEEK 7: MAJOR PROJECT IDEATION, TEAM FORMATION & PLANNING

1. WEEK 7 OVERVIEW

Week Name: Major Project Ideation & Planning

Duration: 5–7 Days

Main Objective

Week 7 marks the beginning of the **major project phase**.

Students will use everything learned in the previous six weeks to identify a meaningful real-world problem, form suitable teams, and plan a larger project that can become part of their portfolio.

The focus is not simply on creating another college project.

The goal is to build a project that demonstrates:

- Technical skills.
 - Problem-solving ability.
 - Teamwork.
 - Research.
 - Documentation.
 - Real-world thinking.
 - Product development.
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2. WEEK 7 GOALS

By the end of Week 7, every project team should:

- Identify a real-world problem.
- Conduct basic problem research.
- Define target users.
- Form a suitable project team.
- Select required technologies.
- Define project features.
- Create an MVP plan.
- Divide the project into tasks.
- Create a GitHub repository.
- Prepare project documentation.
- Present the project proposal.

3. DAY 1 — PROBLEM DISCOVERY

Training Topic

Finding Problems Worth Solving

Students should learn how to identify real problems instead of immediately starting with technology.

Possible Problem Areas

- Education.
- Healthcare.
- Agriculture.
- Campus management.
- Environment.
- Accessibility.
- Transportation.
- Productivity.
- Electronics and automation.
- IoT.
- EV and energy.
- Small businesses.
- Community problems.

Problem Identification Framework

```
Observe Problem
↓
Identify Users
↓
Understand Pain Points
↓
Research Existing Solutions
↓
Identify Gaps
↓
Define Opportunity
```

Student Activity

Teams create a list of possible problems and select one based on:

- Real-world relevance.
- Technical feasibility.
- Available skills.
- Time available.
- Potential impact.

Deliverable

Problem Statement Document

4. DAY 2 — RESEARCH & SOLUTION DESIGN

Training Topic

Research Before Building

Teams research:

- Existing solutions.
- Current alternatives.
- Target users.
- Technology options.
- Required resources.
- Technical limitations.

Project Research Questions

1. What problem are we solving?
2. Who experiences this problem?
3. How is the problem currently solved?
4. What are the limitations of existing solutions?
5. What makes our solution different?
6. Is the project technically feasible?
7. What resources are required?

Deliverable

Research and Solution Document

5. DAY 3 — TEAM FORMATION & ROLE ASSIGNMENT

Teams should be formed based on:

- Skills.
- Domain interests.
- Learning goals.
- Project requirements.

Possible Team Roles

- Project Lead.

- Frontend Developer.
- Backend Developer.
- AI/ML Developer.
- Embedded/IoT Developer.
- UI/UX Designer.
- Research and Documentation.
- Testing and Quality Assurance.

Roles should not permanently restrict students.

Students should be encouraged to learn outside their assigned role.

Team Activity

Each team creates:

- Team name.
- Team members.
- Member roles.
- Communication channel.
- GitHub repository.
- Project workspace.

6. DAY 4 — PROJECT ARCHITECTURE & MVP PLANNING

Training Topic

Design Before Development

Teams define:

- Core features.
- Optional features.
- Technology stack.
- System architecture.
- Database requirements.
- APIs.
- Hardware requirements where applicable.

MVP Definition

Teams divide features into:

MUST HAVE

Essential features required for a working project.

SHOULD HAVE

Important but not essential features.

NICE TO HAVE

Optional future improvements.

Architecture Flow

```
graph TD;
    User --> Frontend["Frontend / Interface"];
    Frontend --> Backend["Backend / Processing"];
    Backend --> Database["Database / Storage"];
    Database --> External["External APIs / Hardware / AI Models"];
```

Deliverable

MVP and Architecture Document

7. DAY 5 — PROJECT PROPOSAL PRESENTATION

Each team presents:

1. Problem.
2. Target users.
3. Research.
4. Existing solutions.
5. Proposed solution.
6. Key features.
7. Technology stack.
8. Team members and roles.
9. MVP.
10. Development timeline.

Mentors provide feedback.

Teams may need to:

- Improve the problem statement.
- Reduce the scope.
- Change the technology.
- Redefine the MVP.

Final Deliverable

Approved Project Proposal

8. WEEK 7 PROJECT DOCUMENTATION

Every major project should contain:

```
project-name/  
├── README.md  
├── docs/  
│   ├── problem-statement.md  
│   ├── research.md  
│   ├── requirements.md  
│   ├── architecture.md  
│   └── project-plan.md  
├── src/  
├── assets/  
├── tests/  
└── CONTRIBUTING.md
```

9. WEEK 7 XP SYSTEM

Activity	XP
Problem research	30 XP
Problem statement	30 XP
Project team formation	20 XP
Project architecture	50 XP
MVP planning	50 XP
Documentation	30 XP
Proposal presentation	75 XP
Week completion	100 XP

10. WEEK 7 COMPLETION CHECKLIST

- ☐ Identify a real-world problem.
 - ☐ Research existing solutions.
 - ☐ Define target users.
 - ☐ Form project team.
 - ☐ Assign initial roles.
 - ☐ Create project repository.
 - ☐ Define project requirements.
 - ☐ Create MVP.
 - ☐ Design basic architecture.
 - ☐ Prepare documentation.
 - ☐ Present the project proposal.
 - ☐ Receive mentor approval.
-

WEEK 7 FINAL OUTCOME

By the end of Week 7, every team should have an approved project with:

- A validated problem statement.
 - Defined target users.
 - Basic research.
 - A suitable team.
 - A defined MVP.
 - A technology stack.
 - A basic system architecture.
 - A project repository.
 - A development plan.
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WEEK 8: MAJOR PROJECT DEVELOPMENT — SPRINT 1

1. WEEK 8 OVERVIEW

Week Name: Major Project Development — Sprint 1

Duration: 5–7 Days

Main Objective

Week 8 begins the actual development of the major project.

Teams will convert their planning into a functional MVP.

The focus should be on completing the **core working functionality**, not building every possible feature.

2. WEEK 8 GOALS

By the end of Week 8, each team should:

- Set up the project environment.
 - Create the project structure.
 - Configure required tools.
 - Set up databases if required.
 - Create the initial frontend/interface.
 - Develop core backend logic where applicable.
 - Integrate required hardware, APIs, or models.
 - Complete the first working version of the MVP.
 - Track tasks and contributions.
 - Conduct a Sprint 1 review.
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3. DAY 1 — PROJECT SETUP

Teams set up:

- GitHub repository.
- Branch strategy.
- Development environment.
- Project structure.
- Required dependencies.
- Database connection.
- Environment configuration.
- Task board.

Recommended GitHub Workflow

```
main
|
├─ develop
|
├─ feature/frontend
├─ feature/backend
├─ feature/database
└─ feature/integration
```

Students should avoid directly pushing unfinished work to the main branch.

4. DAY 2 — CORE DEVELOPMENT

Teams begin working on the most important project features.

Priority:

Core Functionality



Basic User Flow



Data Processing



Database / Storage



Integration

Every team member should have clearly assigned tasks.

5. DAY 3 — INTEGRATION

Teams begin combining individual components.

Examples:

Web Project

Frontend



API



Backend



Database

IoT Project

Sensor



Microcontroller



Communication



Server / Dashboard

AI Project

Dataset
↓
Preprocessing
↓
Model
↓
Prediction
↓
Application Interface

6. DAY 4 — TESTING & BUG FIXING

Teams test:

- Core functionality.
- Inputs and outputs.
- Edge cases.
- User flows.
- Integration points.
- Hardware connections where applicable.

Students document:

- Bugs found.
 - Bugs fixed.
 - Known limitations.
 - Features still pending.
-

7. DAY 5 — SPRINT 1 DEMO

Each team demonstrates its progress.

Sprint Demo Structure

1. Original problem.
2. Project objective.
3. Features completed.
4. Current MVP status.
5. Technology stack.
6. Demonstration.
7. Challenges.
8. Pending work.
9. Next sprint goals.

Mentors provide technical feedback.

8. PROJECT TASK MANAGEMENT

Each project should maintain a task board.

```
BACKLOG
↓
TO DO
↓
IN PROGRESS
↓
CODE REVIEW
↓
TESTING
↓
DONE
```

Each task should contain:

- Task title.
- Description.
- Assigned member.
- Priority.
- Deadline.
- Current status.

9. CONTRIBUTION TRACKING

Noxus should track contributions using:

- Assigned tasks.
- Completed tasks.
- Relevant GitHub commits.
- Pull requests.
- Code reviews.
- Documentation.
- Testing.
- Mentor feedback.

Contribution Formula

```
Contribution Score
=
Task Completion
+
```

Work Quality
+
Collaboration
+
Documentation
+
Review Participation
+
Mentor Evaluation

GitHub commits should act as supporting evidence and not the only measure.

10. WEEK 8 XP SYSTEM

Activity	XP
Development environment setup	25 XP
Complete assigned task	25-75 XP
Pull request merged	25 XP
Integration contribution	50 XP
Testing contribution	30 XP
Documentation	20 XP
Sprint demo	50 XP
Core MVP contribution	100 XP
Week completion	125 XP

11. WEEK 8 STUDENT DASHBOARD

WEEK 8 – MAJOR PROJECT SPRINT 1

Project: [Project Name]

Role: [Student Role]

Assigned Tasks: 5

Completed: 3

In Progress: 1

In Review: 1

Pull Requests: 3

Documentation: Completed

Personal Contribution: 78%

Team MVP Progress: 65%

Week XP: 425 XP

Status: ON TRACK

12. WEEK 8 COMPLETION CHECKLIST

- ☐ Development environment configured.
- ☐ Project structure created.
- ☐ Required dependencies installed.
- ☐ GitHub workflow configured.
- ☐ Tasks assigned.
- ☐ Core features started.
- ☐ Individual components developed.
- ☐ Initial integration completed.
- ☐ Testing performed.
- ☐ Bugs documented.
- ☐ Sprint 1 demo completed.
- ☐ Mentor feedback received.

WEEK 8 FINAL OUTCOME

By the end of Week 8, every team should have:

1. A properly configured project environment.
2. A structured project repository.
3. Active team collaboration.
4. Core features under development.
5. Initial component integration.
6. A partially or fully working MVP.
7. Documented issues and limitations.
8. Sprint progress data.
9. Measurable individual contributions.
10. A clear plan for Sprint 2.

WEEK 7 & WEEK 8 JOURNEY

WEEK 7

Problem Discovery

↓

Research

↓

Team Formation

↓

Solution Design

↓

MVP Planning

↓

Architecture

↓

Project Proposal

↓

Approval

WEEK 8

Project Setup

↓

Task Assignment

↓

Core Development

↓

Component Development

↓

Integration

↓

Testing

↓

Bug Fixing

↓

Sprint 1 Demo

↓

READY FOR WEEK 9 & 10: PROJECT COMPLETION, DEPLOYMENT, PORTFOLIO & PROFESSIONAL READINESS